

Sadly, these extremely long horizons are not relevant for most investors. Only a few institutions—the Catholic Church, the Padmanabhaswamy Temple in Kerala, India (founded in the sixteenth century and possessor of \$22 billion in gold),²⁵ and certain universities such as the University of Bologna (the world’s first, founded in the eleventh century)—have survived for many centuries. But having survived for centuries does not mean these institutions’ planning horizons are centuries-long. The long-term investors of today—pension funds, endowments, family offices, sovereign wealth funds—do not plan 1,000 years ahead.²⁶

Gold may serve valid roles in an investor’s portfolio, but being an inflation hedge is not one of them (at least for horizons less than a century). So why invest in gold?

Gold is often regarded as a safe haven, and Erb and Harvey (2012) show that gold can serve as catastrophe insurance. Tiny increases in the likelihood of disaster raise the gold price substantially. Thus gold could serve as a hedge for disaster risk—which is also one of the main explanations for the high equity-risk premium (see chapter 8).

Ray Dalio, a superstar hedge fund manager who popularized risk parity (which is a special case of mean-variance investing; see chapter 3), makes an argument that gold “is a very under-owned asset” and that if many investors move to increase their holdings of gold to the level reflective of gold’s market capitalization in a world wealth portfolio, the gold price should increase.²⁷ At first glance, this is a silly argument. All the gold in the world—whether in the ground, stored in a vault, or even resting in a shipwreck on the seabed—is already owned by someone. Dalio’s argument can work only if there are market participants who buy to increase their holdings of gold to market capitalization weights and are completely insensitive to the price they pay. That is, some investors must have *perfectly inelastic* demand curves. There may be some institutions like this—some central banks, the growing rich and middle-class populations in emerging markets, and perhaps some pension funds that (mistakenly) believe gold is an inflation hedge. Erb and Harvey (2012) estimate that the market weight of gold as a fraction of total wealth ranges between 2% and 10%, depending on whether you count only the gold held by investors or the entire gold supply, which includes gold yet to be mined and gold held by central banks.

When I look at Panel A of Figure 11.10, I am struck by the very long-term mean reversion exhibited by the real price of gold. The correlation of twenty-year real

²⁵ Jake Halpern, “The Secret of the Temple,” *New Yorker*, April 30, 2012.

²⁶ Perhaps Norway is an exception. The Norwegian sovereign wealth fund pays out approximately 4% of the fund to the Norwegian government each year (the *Handlingsregel*). The Norwegian government fully funded the construction and is meeting a large part of the operational costs, of the Svalbard Global Seed Vault, which seeks to preserve seeds of plants around the world in a remote, secure underground cavern near the North Pole. It is insurance for the world in case of a global catastrophe.

²⁷ Ward, S., “Observing a Bipolar World,” *Barron’s*, March 12, 2011.